Weakening is height-preserving admissible in G3c

The Open Logic Project contributors

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Editorial note. Scope declaration: the author demonstrates the argument through the right-and instance and states the remaining rule cases follow identically; the card declares this uniform-case scope rather than re-deriving each instance.

Editorial scope. One original proposition of the unit's structural admissibility section, counted once. The counted proposition and its author proof are retained.

Allowed standard prerequisites (editorial).

- The calculus G3c, its atomic axioms, and the notion of the height of a proof.

Proposition 0.1. *The rules and of G1c are height-preserving admissible in G3c.*

Proof. We have to show that if $\mathbf{G3c} \vdash \Gamma \Rightarrow \Delta$ with a proof π of height $\operatorname{ht}\pi = n$ then there is a G3c-proof π' of $\Gamma \Rightarrow \Delta, A$ with $\operatorname{ht}\pi' = n$. We prove this by induction on n. If $n = 0$, then π is just the axiom $\Gamma \Rightarrow \Delta$ (i.e., some atomic C occurs in both Γ and Δ), and so is $\Gamma \Rightarrow \Delta, A$. If $n = \operatorname{ht}\pi > 0$, the proof π has a last inference. We distinguish cases according to which rule was used. We give just one example: suppose the last inference was . Then $\Delta = \Delta', B \wedge C$ and the immediate sub-proofs π_1 and π_2 of the last inference have end-sequents $\Gamma \Rightarrow \Delta', B$ and $\Gamma \Rightarrow \Delta', C$, respectively. In other words, π looks like this:

$$\dfrac{\begin{array}{cc}\vdots\,\pi_1 & \vdots\,\pi_2 \\ \Gamma\Delta', B & \Gamma\Delta', C\end{array}}{\Gamma\Delta', B \wedge C}$$

We also have that $n = \max(\operatorname{ht}\pi_1, \operatorname{ht}\pi_2) + 1$. So $\operatorname{ht}\pi_1 < n$ and $\operatorname{ht}\pi_2 < n$, and the inductive hypothesis applies: there are proofs π_1' and π_2' of $\Gamma \Rightarrow \Delta', B, A$ and $\Gamma \Rightarrow \Delta', C, A$. Using the rule, we obtain proof π' of $\Gamma \Rightarrow \Delta', B \wedge C, A$:

$$\dfrac{\begin{array}{cc}\vdots\,\pi_1' & \vdots\,\pi_2' \\ \Gamma\Delta', B, A & \Gamma\Delta', C, A\end{array}}{\Gamma\Delta', B \wedge C, A}$$

We have $\operatorname{ht}\pi' = \max(\operatorname{ht}\pi_1', \operatorname{ht}\pi_2') + 1 = \max(\operatorname{ht}\pi_1, \operatorname{ht}\pi_2) + 1 = \operatorname{ht}\pi$. □